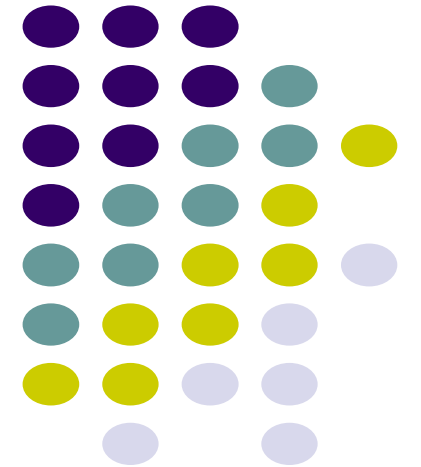


Resins



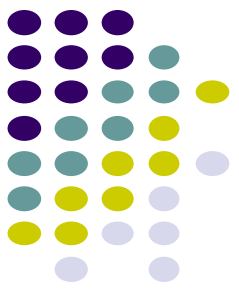
Objective



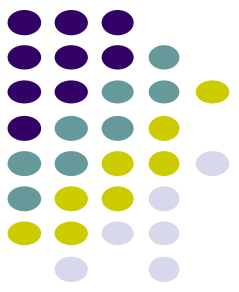
At the end of this chapter students will be able to:

- define and classify resins
- identify resins
- mention their use
- describe their physical and chemical characteristics

Introduction

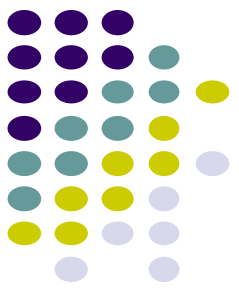


- Resins are solid or semi-solid amorphous products of complex chemical nature.
- They are considered as end products of metabolism, and are believed to arise by oxidation of polymerized terpenoids metabolites forming complex mixtures
- They are simply extractions of plant material, are taken either from
 - the whole plant or
 - from specific parts of the plant (bark of trees, flowers of herbs, and buds of shrubs)



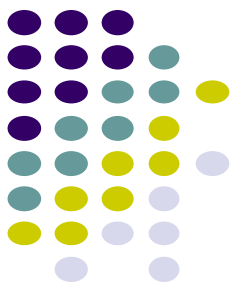
Occurrence

- Resins can be obtained from plant and animal sources are known as natural sources resins.
- They are found inside plants or are exuded by plants, such as sap, latex, or mucilage forms.
- It can be either in the internal part or on the surface of different parts of the plant.
- They are obtained in different parts of plants
 - Resin Cell – Ginger
 - Glandular Hair – Cannabis
 - Schizogenous or Schizolysigenous Duct or Cavities – Pinewood
 - Induced at a Site of Injury / Incision - Benzoin



- Some examples of plant source resins are *Asafoetida*, *Myrrh*, *Balsam*, *Benzoin*, *Ginger*, *Colophony*, *Jalap*, *Podophyllum*, etc.
- Some examples of animal source resins are *Shellac* or *Lac*.
- Sometimes animal and plants resins are obtained from fossils, they are known as fossilized resin. Example: *Amber* which is fossilized tree resin, often known as *Copal*.

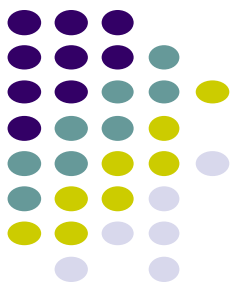




Properties

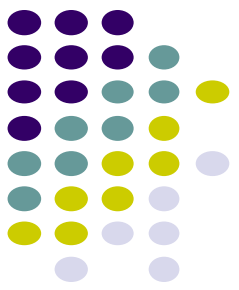
- These are transparent or translucent solid or semisolid.
- They are brittle solid.
- They are fusible and flammable organic substances.
- The specific gravity of Resins is more than water. Therefore, these are heavier than water
- They are soluble in volatile oils, ether, and alcohol.
- They generally become soft at heating. On further heating, Resins will be melted
- These compounds are highly enriched with carbon.
- On storage, resins undergo a slow oxidation process in the atmosphere and become dark in color.

Chemistry



- The resin produced by most plants is a viscous liquid, composed mainly of volatile fluid terpenes, with lesser components of dissolved non-volatile solids which make resin thick and sticky
- The common terpenes in resin are the bicyclic terpenes alpha and beta-pinene and sabinene, the monocyclic terpenes limonene and terpinolene, and smaller amounts of the tricyclic sesquiterpenes.
- Resins are enriched with carbon, deprived of nitrogen and contain oxygen.
- Resins generally form soap when boiled with alkali.
- They are associated with volatile oils (oleoresins), with gum (gum resins), or with oil and gum (oleo-gum resins).
- Electrically they are non-conductive masses

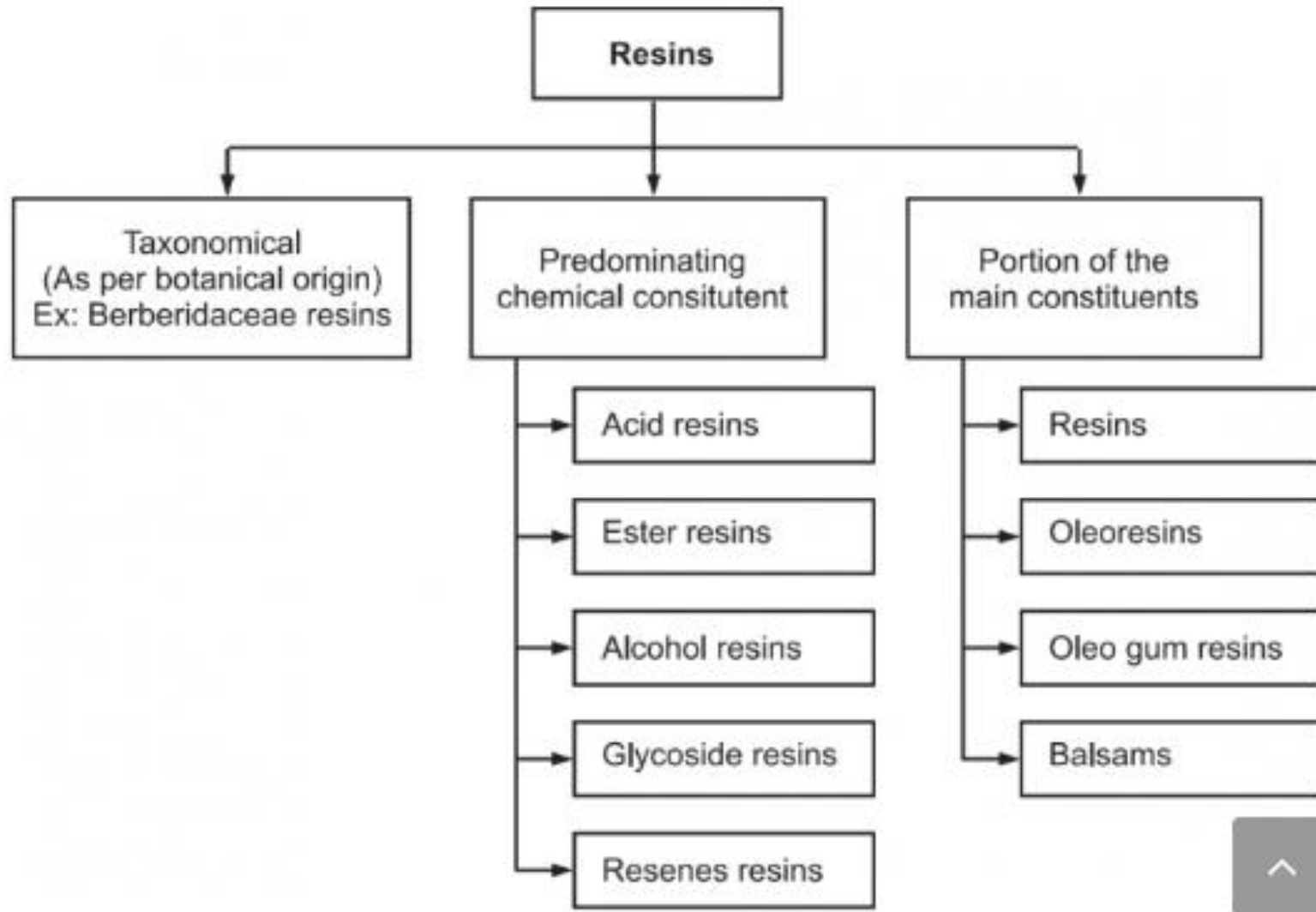
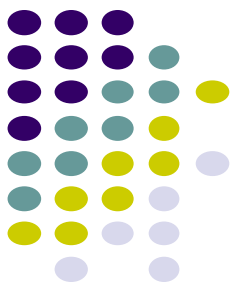
How is Resin Made?



Based on their formation:

- Physiological Resin: These types of Resins are formed by the normal metabolism process. Example - cannabis, podophyllum, and ginger.
- Pathological Resin: These types of Resin are formed by the result of the wound, injury, or abnormal circumstances. Example - benzoin, asafoetida, and guggul.

Classification of Resins

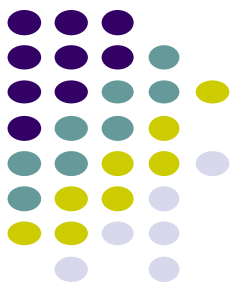




Taxonomical Classification

The resins are grouped according to their botanical origin exclusively, such as:

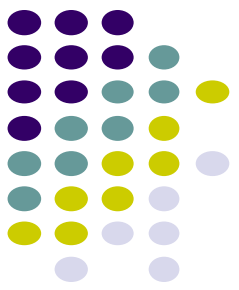
- **Coniferous Resins:** e.g., Colophony, Sandarac;
- **Berderidaceae Resins:** e.g., Podopyllum and
- **Zygophyllaceae Resins:** e.g., Guaiacum
- In this particular instance, it has been observed that the resins that usually occur in plants of the same natural order, may exhibit more or less related characteristics features.



Resin combination

Classification of Resin is based on the nature of occurrence with other secondary metabolites.

- **Oleoresin-** These are naturally occurring Resin, which is a mixture of Resin and volatile oil. possesses an essence due to volatile oils.
 - Eg. capsicum, ginger, and copaiba.
- **Gum Resin-** These types of Resins are associated with the gum.
 - Due to solubility in water, gums can be easily separated out from resin by dissolving the gum in water
Eg. colophony and cannabis. Ammoniacum is an example of natural gum resin.

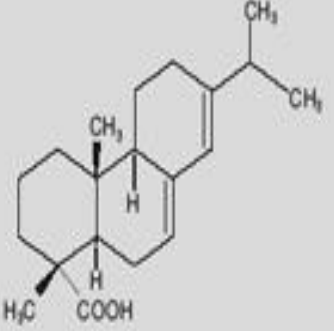
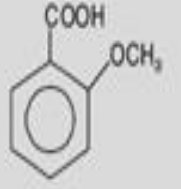
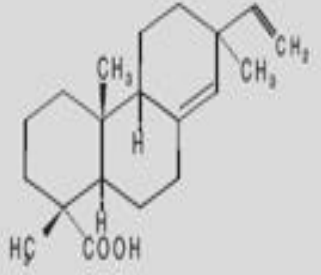


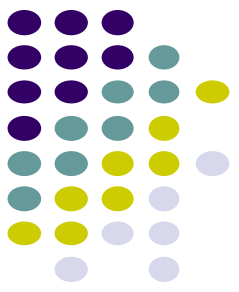
- **Oleo Gum Resins-** These types of Resins are a mixture of volatile oil, gum, and Resin.
 - Eg turmeric, asafoetida, and myrrh.
- **Balsams-** These types of Resin are a mixture of benzoic acid and cinnamic acid or esters of these acids.
 - It can occur in free or combined form.
 - Eg. benzoin, tolu balsam, Peru balsam.
- **Glyco Resin-** This type of Resin occurs in combination with sugar.
 - Eg. jalap and podophyllum

Chemical classification

Classification on the basis of chemical constituents:

- **Acid resin-** are mixtures of large proportion of diterpenoid oxyacids, carboxylic acids and phenolic acids
 - Found in free states or as the esters derivatives.
 - They combine with alkali and their metallic salts are termed as resonates.
 - With aqueous solution of alkali they form soap like solution or colloidal suspension.
 - Eg: abietic acid (colophony), copaivic acid and oxycopaivic acid (copiba), primaric acid (fankicense)

S. No.	Name	Chemical Structure	Source(s)
1	Abietic Acid		Colophony, Rosin
2	Copaivic Acid and Oxycopaivic Acid	—	Copaiba
3	Guaiaconic Acid		Guaic
4	Pimaric Acid		Burgandy Pitch, Fankicense
5	Sandarcolic Acid	—	Sandarac
6	Commiphoric Acid	—	Myrrh

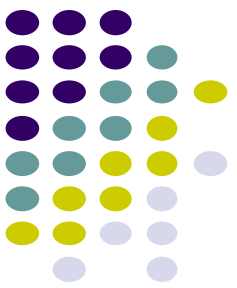


Resin alcohol-

- These are complex alcohols of high molecular weights
 - present in free state or as ester of simple aromatic acids like benzoic acid, cinnamic acid, and salicylic acid
 - Resin alcohols are further subdivided into:
 - I. Resinotannols (resin phenols)
 - have tannin like properties and form blue color with ferric chloride
Eg: Aloeresinotannol from aloe
 - II. Resinols
 - are high molecular weight non phenolic alcohols which do not give tannin reaction with iron salts
Eg: Benzoresinol from benzoin



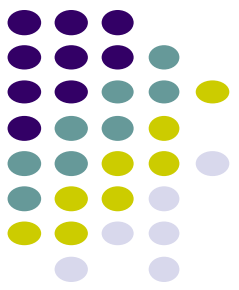
- **Ester resins-** these are esters of resin alcohol or resinotannol combined with resin acid or balsamic acid.
Eg: cinnamyl cinnamate from storax
- **Resenes-** these re the neutral and inert substances as they do not contain characteristic functional group
Eg: Asafoetida contains drug about 50% of asaresene B.



Uses

- In bronchia asthma (Cannabis)
- It is used as an anticancer drug (podophyllum)
- They are local irritant and act as local cathartics (Jalap and Ipomea) property.
- It is used as a stimulant or diuretic agent.
- It is used as an anti-inflammatory property.
- Used externally as mild antiseptic in the form of tinctures (Benzoin)
- Resins are used as a carminative agent.
- It is used as an expectorant.
- These are used as flavoring agents.
- It is also used for bow treatments for instruments like cellos and violins

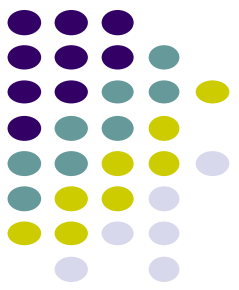
Preparation of Resins



- So far, no general method has either been suggested or proposed for the preparation of resins.
- In fact, there are two categories of resinous products, namely: (a) Natural Resins; and (b) Prepared Resins

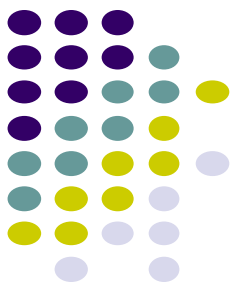
A. Natural Resins:

- These resins usually formed as the exudates from various plants obtained either normally or as a result of pathogenic conditions (i.e., by causing artificial punctures), such as: mastic, sandarac.
- These are also obtained by causing deep incisions or cuts in the trunk of the plant, for instance: turpentine.
- They may also be procured by hammering and scorching, such as: balsam of Peru.

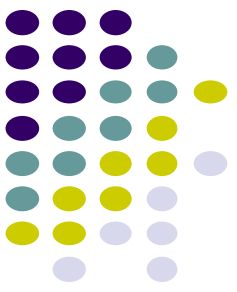


B. Prepared Resins:

- The resins obtained here are by different methods as described below:
- (i) The crude drug containing resins is powdered and extracted with ethanol several times till complete exhaustion takes place. The combined alcoholic extract is either, evaporated on water-bath slowly in a fuming cup-board or poured slowly into cold distilled water.
- The precipitated resin is collected, washed with cold water and dried carefully under shade or in a vacuum desiccator.
- Examples: Podophyllum; Scammony and Jalap



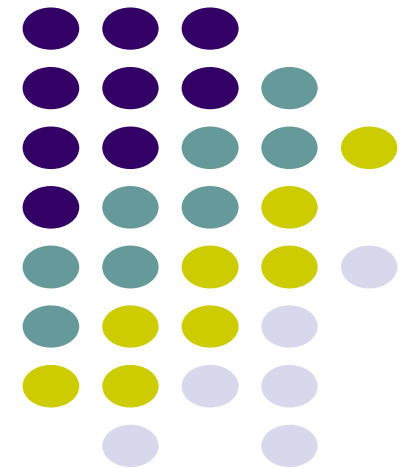
- (ii) In the case of alco-resins, organic solvents with lower boiling points are normally employed e.g., solvent ether (bp 37°C); acetone (bp 56.5°C), for their extraction. However, the volatile oil fraction can be removed conveniently through distillation under vacuo.
- (iii) In the instance of gum-resins, the resin is aptly extracted with 95% (v/v) ethanol while leaving the insoluble gum residue in the flask (or soxhlet thimble).

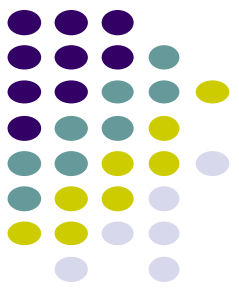


Extraction and isolation

- Pharmaceutical resins are obtained from the plants and animals by one of the following methods:
- By extraction with alcohol and precipitation with water, e.g. jalap, podophyllum,...etc.
- By distillation for separation of oil, e.g. copaiba, colophony,...etc.
- By heating the plant part, e.g. guaiacum.
- As plant exudates by incisions, e.g. myrrh, asafoetida, balsams,...etc.
- By collecting fossil resins, e.g. copal, kauri,...etc.
- By processing the encrustations i.e. shellac

Drugs Containing Resins

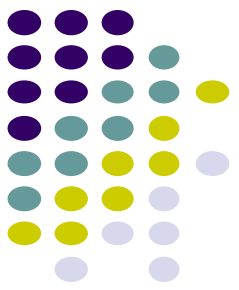




1. Podophyllum

- The term is derived from ancient Greek word means 'foot leaf'
 - also known as Mayapple because of its fruits ripen in spring
- The genus Podophyllum is generally represented by two species:
 - *P. hexandrum* - Himalayan Mayapple
 - *P. peltatum* - American Mayapple
- *P. peltatum* contains lower amounts of resin in comparison to *P. hexandrum*

Podophyllum Cont...



P. hexandrum



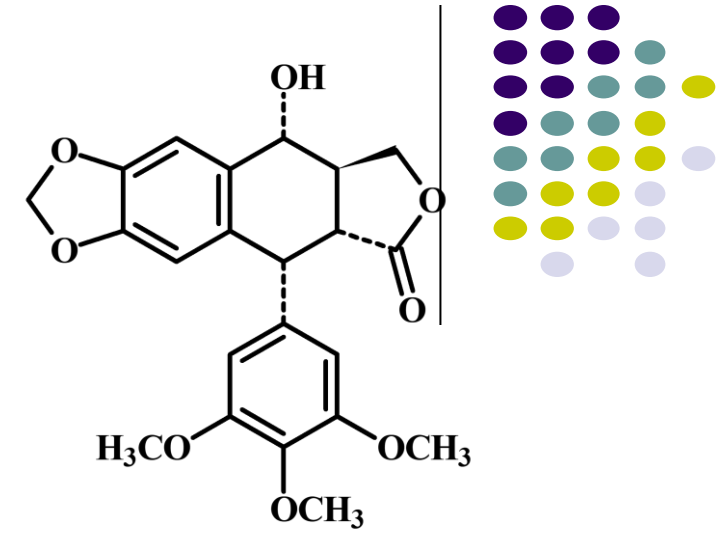
P. peltatum



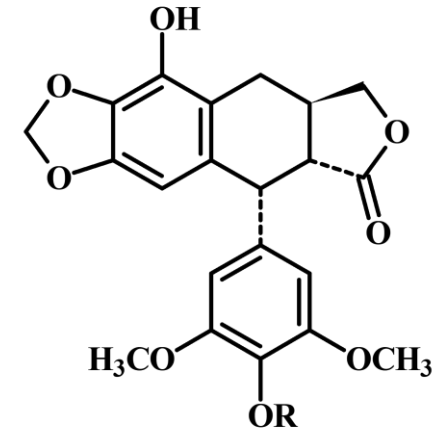
- Podophyllum is an endangered but high value medicinal plant from temperate and cold climatic zones of the globe
- It has wide scope of organized scientific study on its different aspects

Uses

- Podophyllin (a crude resin)
 - vermifuge and emetic
 - purgative
 - antitumour properties
- Podophyllotoxin,
 - blocks polymerization of tubules into microtubules, leading to an arrest of mitosis and thus resulting in destruction of the warts.
- developed to be anticancer agents

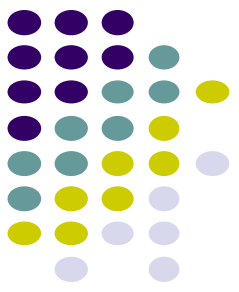


Podophyllotoxin



β -Peltatin, R = CH₃

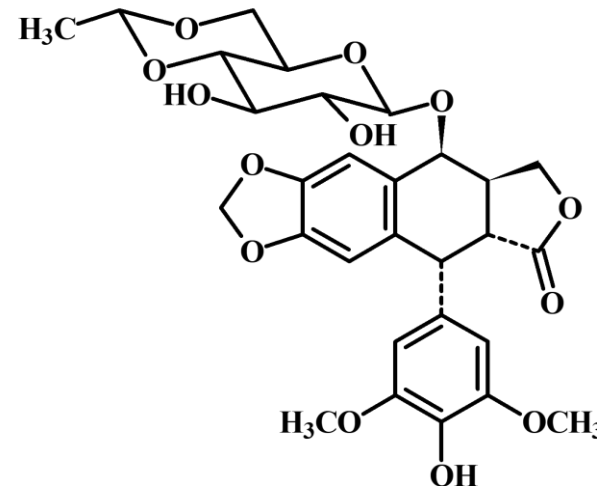
α -Peltatin, R = H



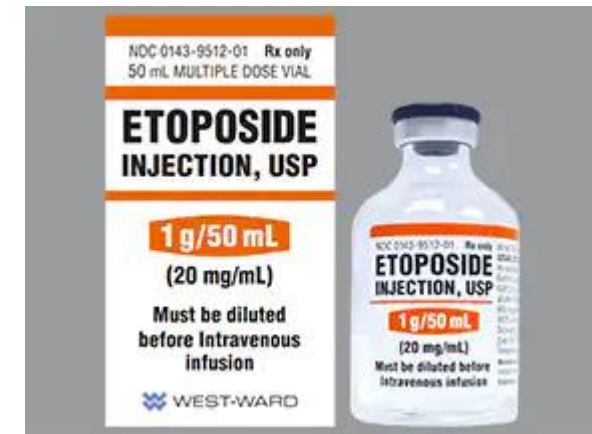
Derivatives of Podophylotoxin

1. Etoposide

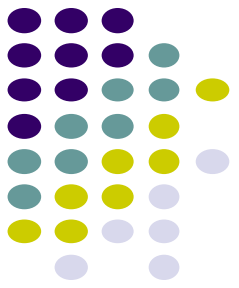
- Lignan derivative obtained semi-synthetically from podophyllotoxin
- Used in the treatment of small-cell lung cancer, testicular cancer, lymphomas and leukaemias



Hiwot A. (B.Pharm, MSc.), August, 2022

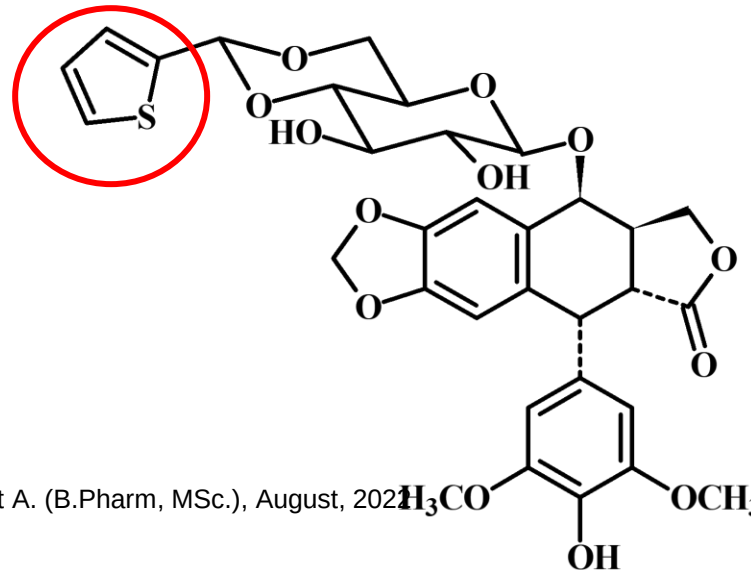


Derivatives Cont...



2. Teniposide

- Thenylidene derivative
- Used in the treatment of paediatric cancer: lymphocytic leukaemia, brain tumours



2. Ginger

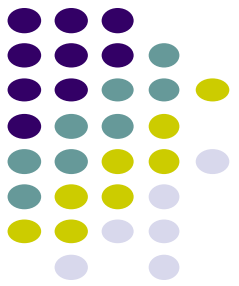
Geographical sources

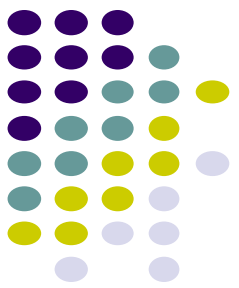
- Jamaica
- China
- India
- Africa



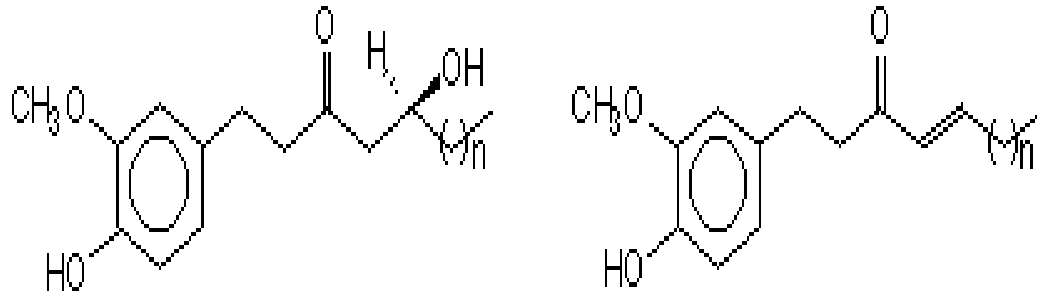
Constituents

- **Volatile oils (1 – 2%)**
 - Camphene
 - Cineole
 - Citral
 - Borneol
 - Sesquiterpene HC's
 - Zingiberene & Zingiberol (Sesquiterpene alcohol)
- **Resins**
 - Gingerols – pungent component – Anti-inflammatory
 - Shogaols – increases bile secretion
- **Starch**
- **Mucilage**



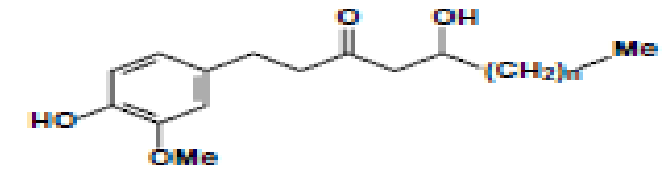
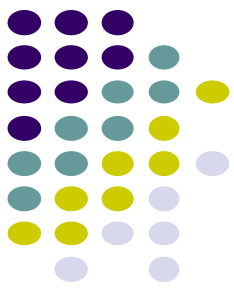


Chemical Structures

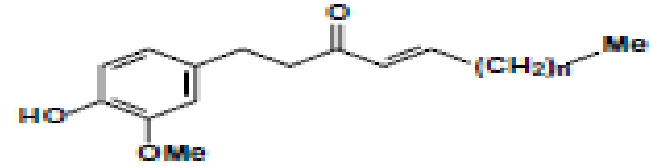


	n				n		
[6] Gingerol	4	C ₁₇ H ₂₆ O ₄	294.3	[6] Shogaol	4	C ₁₇ H ₂₄ O ₃	276.3
[8] Gingerol	6	C ₁₉ H ₃₀ O ₄	322.7	[8] Shogaol	6	C ₁₉ H ₂₈ O ₃	304.4
[10] Gingerol	8	C ₂₁ H ₃₄ O ₄	350.2	[10] Shogaol	8	C ₂₁ H ₃₂ O ₃	332.1

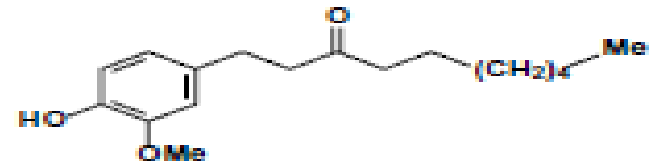
- Gingerols are a homologous series of phenols
 - The most abundant is [6]-gingerol
- The pungency of dry ginger mainly results from shogaols which are dehydrated forms of gingerols



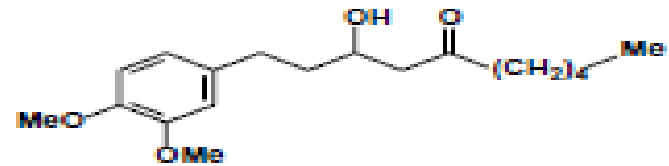
n = 4	[6] - gingerol	1
n = 2	[4] - gingerol	3
n = 5	[7] - gingerol	4
n = 6	[8] - gingerol	5
n = 8	[10] - gingerol	6



n = 4	[6] - shogaol	2
n = 2	[4] - shogaol	7
n = 6	[8] - shogaol	8
n = 8	[10] - shogaol	9
n = 10	[12] - shogaol	10



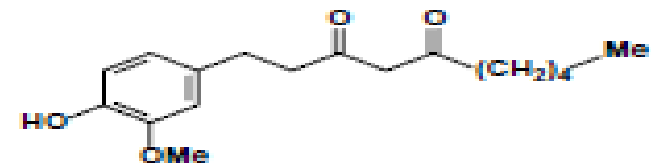
[6] - paradol 11



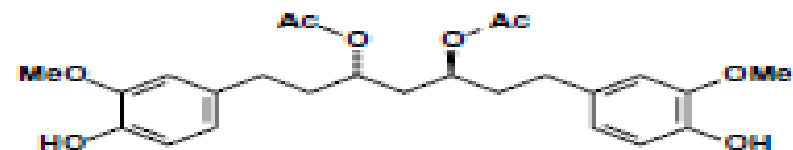
methyl [6] - isogingerol 12



[6] - isoshogaol 13



[6] - gingerdione 14



(3*S*,5*S*) - 3,5 - diacetoxy - 1,7 -
bis (4-hydroxy-3-methoxy) -
heptane 15

Uses and Actions

- **Actions**

- Carminative
- Stimulant
- Anti-emetic
- Anti-bacterial & Anti-fungal
- Gingerol – pungent component – Anti-inflammatory
- Shogaols– increases bile secretion & enhanced GIT activity

- **Uses**

- Motion Sickness
- Morning Sickness



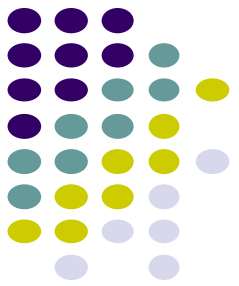
3. Capsicum

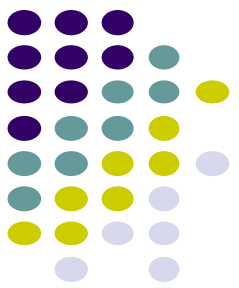
- is the dried ripe fruit of *Capsicum frutescens*, known in commerce African chilies, *C. annum*, known in commerce Tabasco pepper



Chemical Constituents

- capsaicinoids
- volatile oils (1.5%)
- fixed oil
- carotenoids
- ascorbic acid



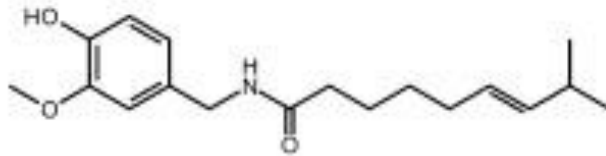


Capsaicinoids

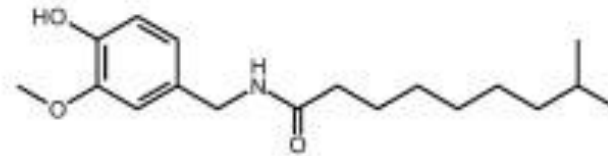
- Capsaicinoids are considered the active ingredients of OC.
- Capsaicinoids are isolated through a solvent extraction of the dried, ripened fruit of chili peppers then dried, and compounded together
- The final oleoresin contains several branched-chain alkyl vanillylamides, in addition to capsaicin, the major component in OC.
- Cause the spicy flavor (pungency) of chili pepper fruit
- The predominant capsaicinoid components of OC are capsaicin (70%), dihydrocapsaicin (20%), norhydrocapsaicin (7%), homocapsaicin (1%), and homodihydrocapsaicin (1%)

Capsaicinoids

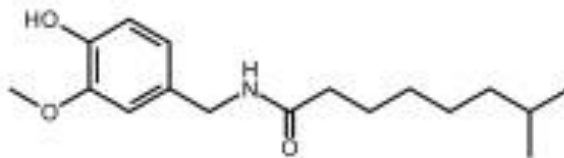
Compounds Found in Oleoresin Capsicum:



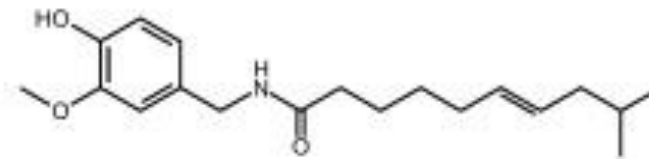
Capsaicin
Trans-8-methyl-N-vanillyl-6-nonenamide



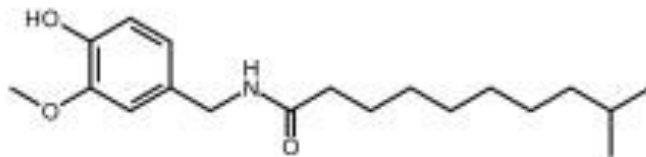
Dihydrocapsaicin
8-Methyl-N-vanillyl-nonamide



Nordihydrocapsaicin
7-Methyl-N-vanillyl-octamide

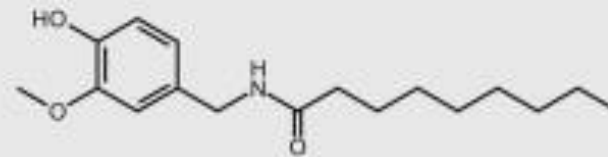


Homocapsaicin
Trans-9-methyl-N-vanillyl-7-decenamide

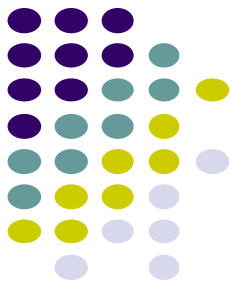


Homodihydrocapsaicin
9-Methyl-N-vanillyl-decamide

Pseudocapsaicin:



Pelargonic Acid Vanillylamide(PAVA)
N-[(4-hydroxy-3-methoxyphenyl)methyl]



Uses

- culinary importance
- used medicinally in creams to counter neuralgia and in other topical pain-relieving preparations
- The initial burning effect of capsaicin is found to affect the pain receptors, making them less sensitive
- antioxidant, anticarcinogenic , promotion of energy metabolism and suppression of fat accumulation , and anti-inflammatory



Myrrh

- oleo-gum resin exudates obtained from several species in the genus *Commiphora*
- Mainly harvested from *Commiphora myrrha*, a low growing, desert tree that exudes a viscous yellow gum, which hardens into a reddish-brown resin.
- It contains :
 - 57–61% water-soluble gum
 - 7–17% volatile oils
 - 25–40% alcohol-soluble resins and
 - 3–4% impurities
- The name myrrh is derived from the Arabic and Hebrew word mur, which means bitter



Table: Occurrence of the different species of *Commiphora*

Species	Country
<i>C. molmol</i> Engler	Ethiopia, Somalia, Arabia
<i>C. mukul</i> Engler	Somalia, India
<i>C. guidotti</i> Chiov.	Somalia
<i>C. abyssinica</i> Engler	China, East Africa, Ethiopia
<i>C. incisa</i> Chiov.	India, East Africa, Ethiopia
<i>C. pyracanthoides</i> Engler	East Africa
<i>C. rostrata</i> Diels	Arabia
<i>C. gileadensis</i> Engler	Djibouti, Ethiopia, Kenya, Somalia, Sudan
<i>C. wightii</i> Engler	Pakistan, India
<i>C. guillauminiperr</i> Engler	Sudan, Kenya
<i>C. erythraea</i> (Var.) Engler	India, Somalia
<i>C. opobalsamum</i> Engler	Near Cairo "at mataria"

Chemistry

- The alcohol-soluble resins of myrrh¹
 - commiphoric acids
 - commiphorinic acid
 - heeraboresene
 - heerabomyrrhols
 - commiferin.....

GOLD, FRANKINCENSE & MYRRH

These three substances are brought as gifts by the Three Wise Men in the Christmas story - in this graphic, we examine the uses and chemistry behind them.



Gold

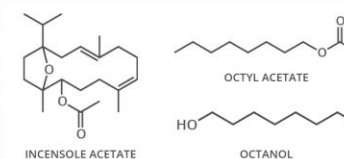


Electron Configuration: $[\text{Xe}]4f^{14}5d^{10}6s^1$

Gold's structure consists of a lattice of cations with a 'sea' of delocalised electrons. Its golden colour is due to it having the correct energy difference between its 5d and 6s electron subshells to absorb photons of blue light.



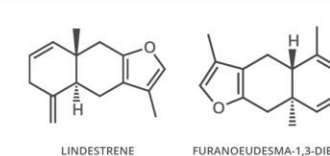
Frankincense



Frankincense resin is used as incense, with its smell the result of various compounds produced when it is burned. These include those shown above, as well as several terpenoid compounds. Research suggests that the compound incensole acetate could help alleviate anxiety & depression.



Myrrh



Myrrh resin, like frankincense, can be used in incense & perfumes, though it also has medicinal applications. Its aroma comes from a family of compounds called furanosesquiterpenes. Animal studies have suggested some compounds in myrrh can have an analgesic effect.

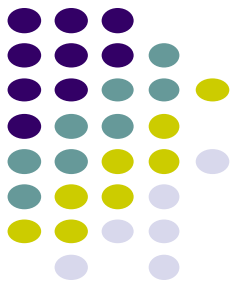
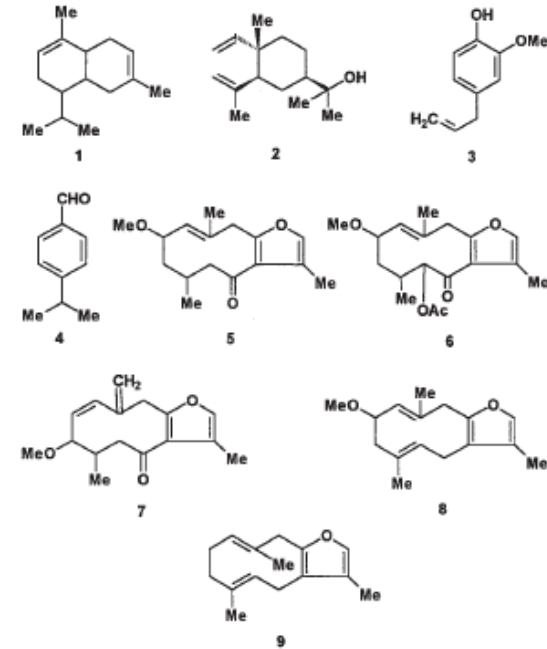


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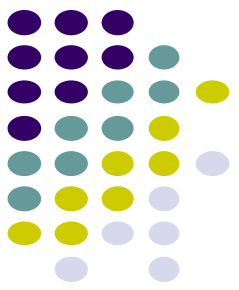


Chemistry Cont...

- The volatile oil fraction contains:
 - Terpenes, elemol
 - cinnamaldehyde
 - cuminaldehyde
 - cumicalcohol
 - eugenol, heerabolene,
 - limonine, dipentene, pinene....
- Its characteristic odor is due to the furanosesquiterpenes
- The water-soluble gum or mucilage fraction is composed mainly of acidic polysaccharide with galactose, xylose, 4-O-methyl-glucuronic acid and arabinose

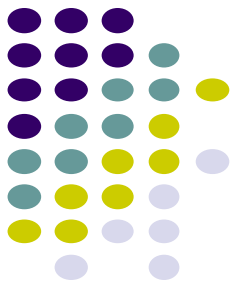
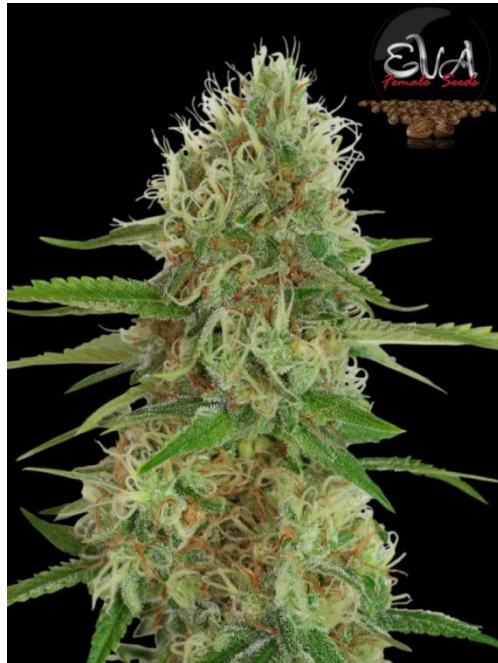


Uses



- food, flavor, liqueur and beverage, cosmetics, perfumery and others.
- frankincense and myrrh have several local applications in medicinal, hygienic, and insecticide areas that could be developed through research
- Myrrh has been shown anti-inflammatory, antipyretic and antimicrobial activity that work in two ways:
 - it stimulates the production of white blood corpuscles (with their anti-pathogenic actions)
 - it has a direct anti-microbial effect

4. Cannabis

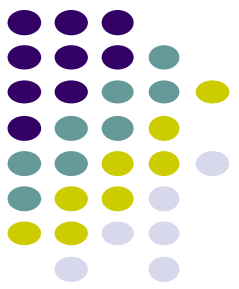




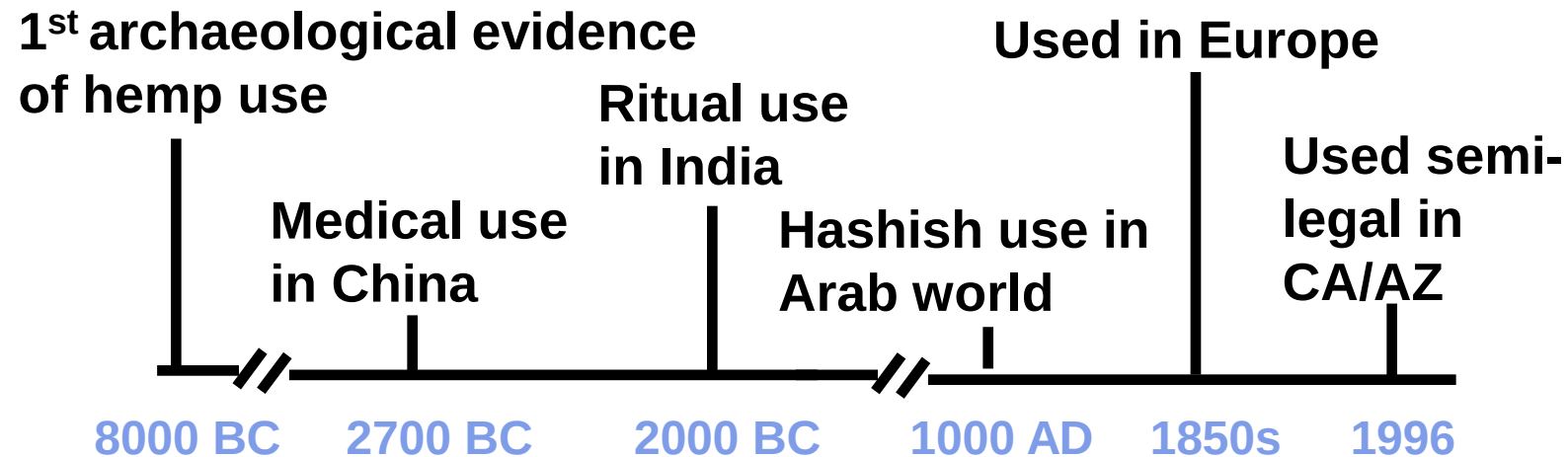
Cannabis

- is the general name given to a variety of substances prepared from the plant *Cannabis sativa*
 - In small doses, it is classified as a depressant
 - In larger doses, it can be an hallucinogen
- The dried plant, resin or oil is commonly smoked for recreational purposes
- can also be ingested by cooking it in food
- is also called 'gunga', 'dope', 'hash', 'weed', 'hemp', 'grass' and 'mull'
- is normally a dioecious plant

History



- Cannabis is among the very oldest of economic plants, providing fiber, edible seed, and drug resin



Why do people use cannabis?

To experience euphoria or perceptual distortions because various mental and physical effects.

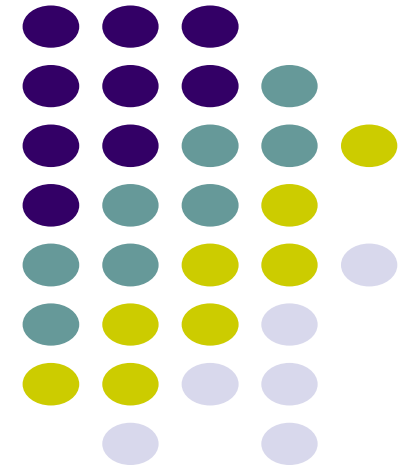
To relieve mental symptoms

- negative symptoms of schizophrenia
- anxiety

To relieve physical symptoms

- eg nausea, anorexia, cancer & HIV

Medical cannabis: the use of cannabis to treat disease or improve symptoms.





Forms and Preparations

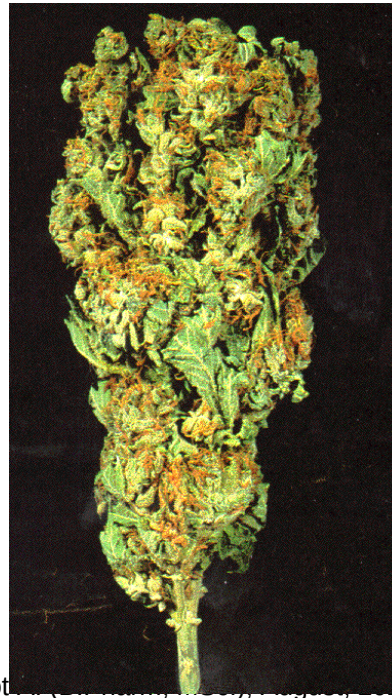
Marijuana

- the dried leaves and flowers (buds) of the cannabis plant that are smoked in a joint or a bong.

- most common form

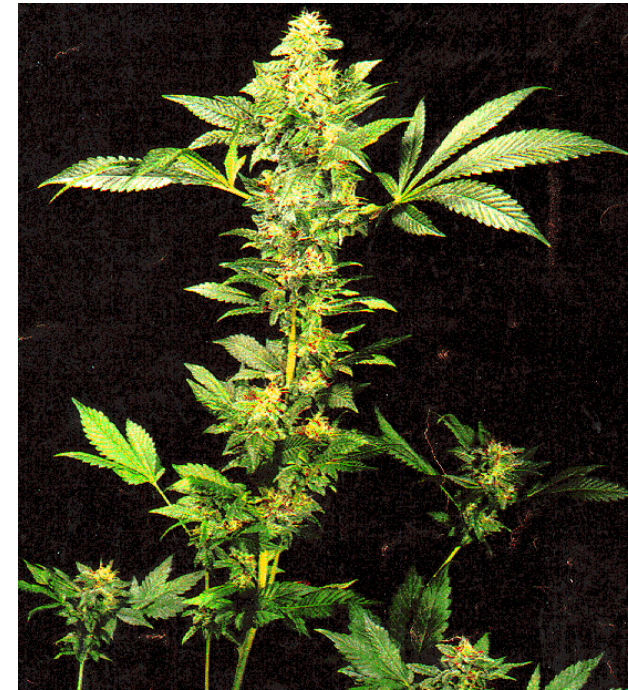


“Bubble Gum”



Hiwot (cannabis), August 2012

“Big Bud”



“Dutch Northern Lights”



Forms and Preparations

- **Hashish**
 - dried resin from top of female plant
 - that is usually mixed with tobacco and smoked or added to foods
 - THC usually 2-5%, but up to 15%
- **Hash Oil**
 - organic extraction from hashish
 - THC usually ~ 10-20% up to 70%





Effects on Behavior

Low - moderate doses

- disinhibition, relaxation, drowsiness
- feeling of well being, exhilaration, euphoria
- sensory - perceptual changes
- recent memory impairment
- balance/stability impaired
- decreased muscle strength, small tremor
- poor on complex motor tasks (e.g., driving)

The Long Term Effects of Cannabis

(when smoked with tobacco the risks are increased)

Eyes

- Sleepy looking
- bloodshot eyes with dilated pupils

Respiratory System

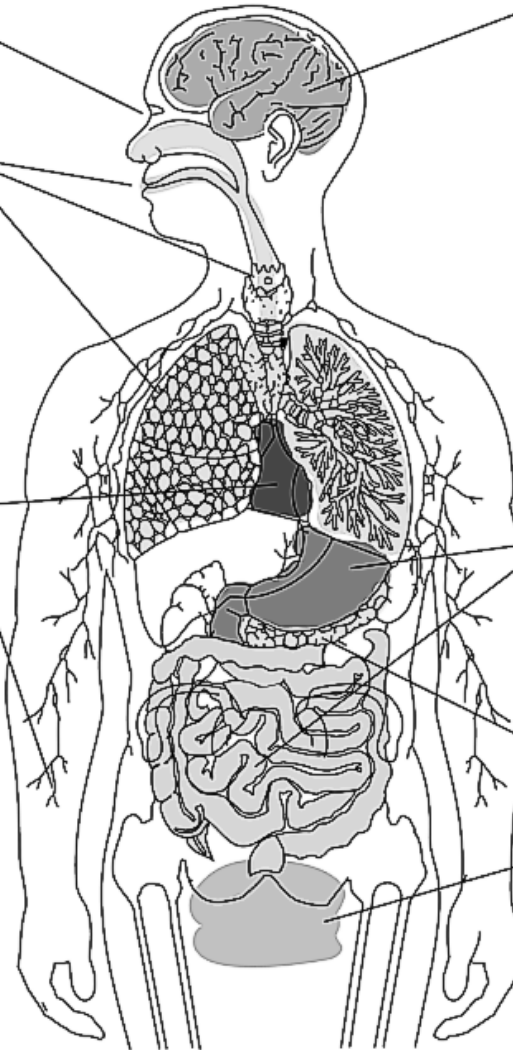
- increased risk of chest infections
- increased risk of lung, mouth & throat cancer
- chronic bronchitis, and emphysema (damage to the air sacs in the lungs)

Circulatory System

- increased heart rate and blood pressure
- increased risk of heart failure
- poor circulation - blood flow (arms & legs)

Pregnancy and Babies

- REGULAR USE DURING PREGNANCY CAN CAUSE:
- lowered birth weight
 - fetal abnormalities
 - small head, irritability, poor growth and development



Central Nervous System (brain and spinal cord)

- psychological disturbances
- exacerbation of schizophrenia and psychosis
- short term memory loss
- imagining and seeing things
- impaired ability to learn new things or do complex tasks
- sleep disturbances
- slower reactions
- anxiety or panic attacks
- loss of motivation
- poor concentration
- change in personality

Stomach and Intestines

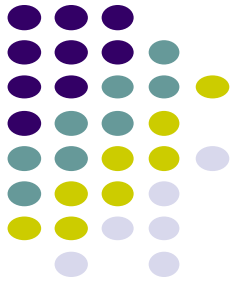
- Lining becomes tender
- bleeding
- ulcers
- may lead to cancer

Pancreas

- Increased risk of cancer

Reproductive System Male and Female

- decreased sperm count and movement
- lowered sex drive
- egg damage
- irregular menstrual cycle and altered hormone levels
- impaired fertility





Effects on behavior

High doses

- pseudohallucinations
- synesthesias
- impaired judgement, reaction time
- pronounced motor impairment
- increasingly disorganized thoughts, confusion, paranoia, agitation

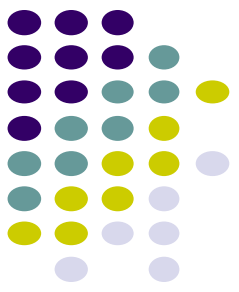
Not lethal even at very high doses

Pharmacokinetics



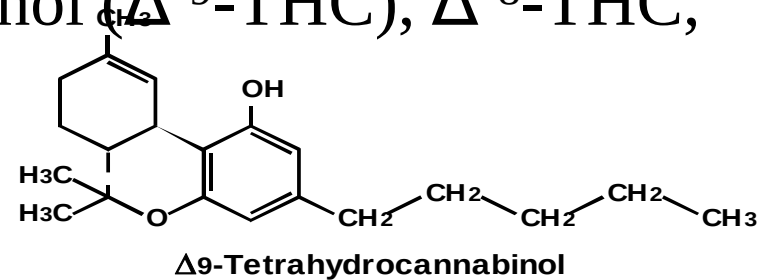
Absorption

- very lipid soluble
- good absorption if smoked (20-37%)
- rapid peak
- slow absorption with oral



Cannabinoids

- a group of terpenophenolics compounds that are pharmacologically active compounds
- over 65 (delta-9-tetrahydrocannabinol (Δ^9 -THC), Δ^8 -THC, cannabinalol, etc.)

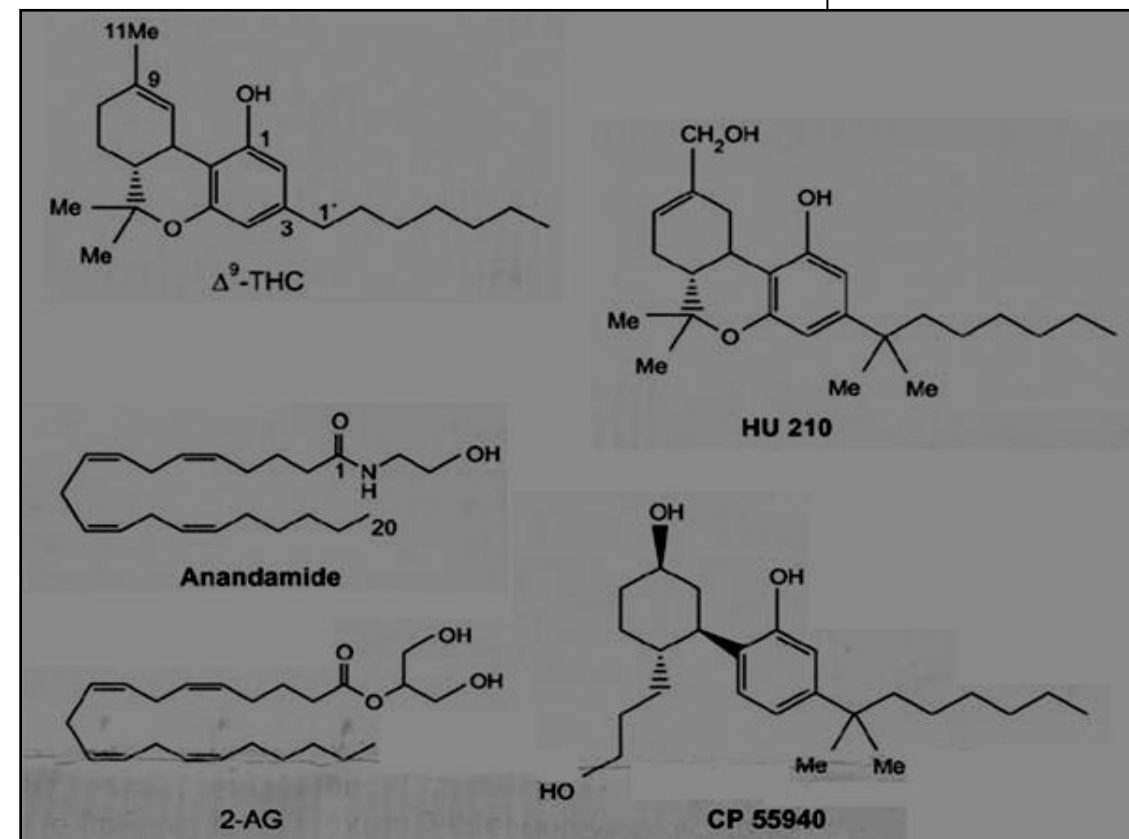


- The FDA has not approved the cannabis plant for any medical use, however it approved several drugs that contain individual cannabinoids.
- Epidiolex, which contains a purified form of CBD derived from cannabis, was approved for the treatment of seizures associated with lennox-gastaut syndrome

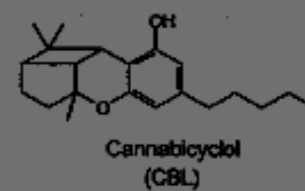
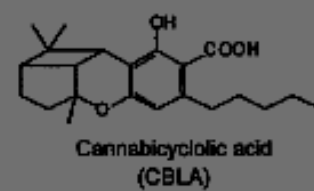
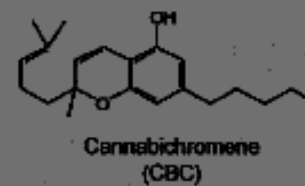
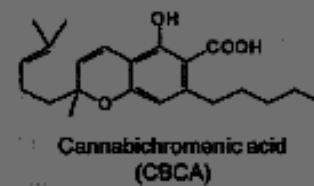
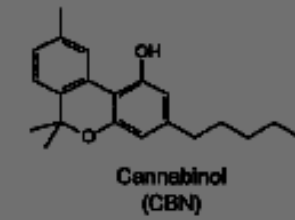
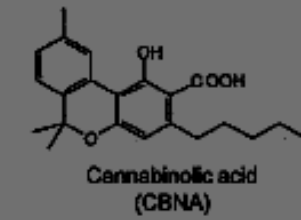
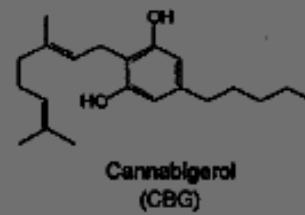
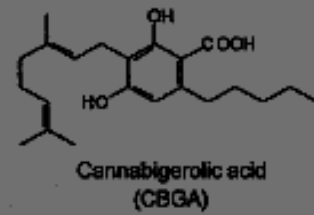
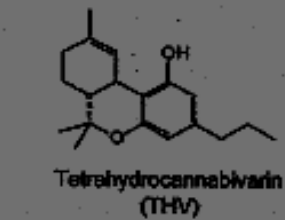
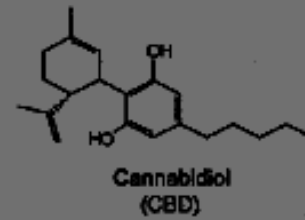
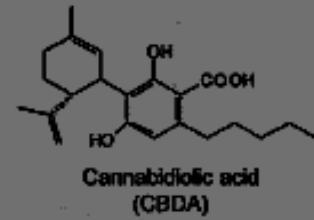
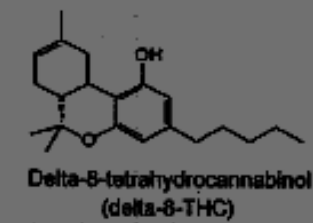
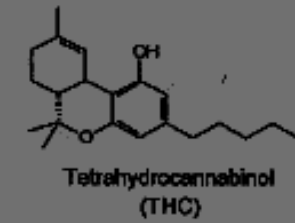
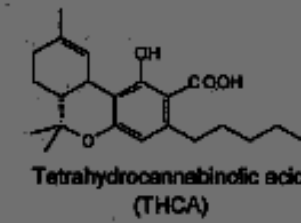
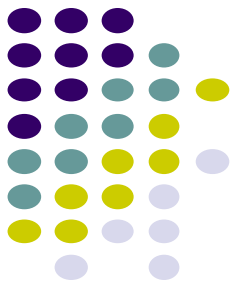
Some Common Cannabinoids

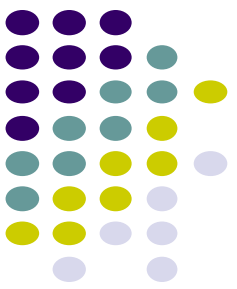


In the plant:	In your body (endogenous):	Synthetic:
Δ^9 – THC	2-arichidonyl glycerol (2 – AG)	CP 55,940
Δ^8 – THC	anandamide	HU – 210
cannabidiol and cannabinol		



Structure of THC and synthetic analogs

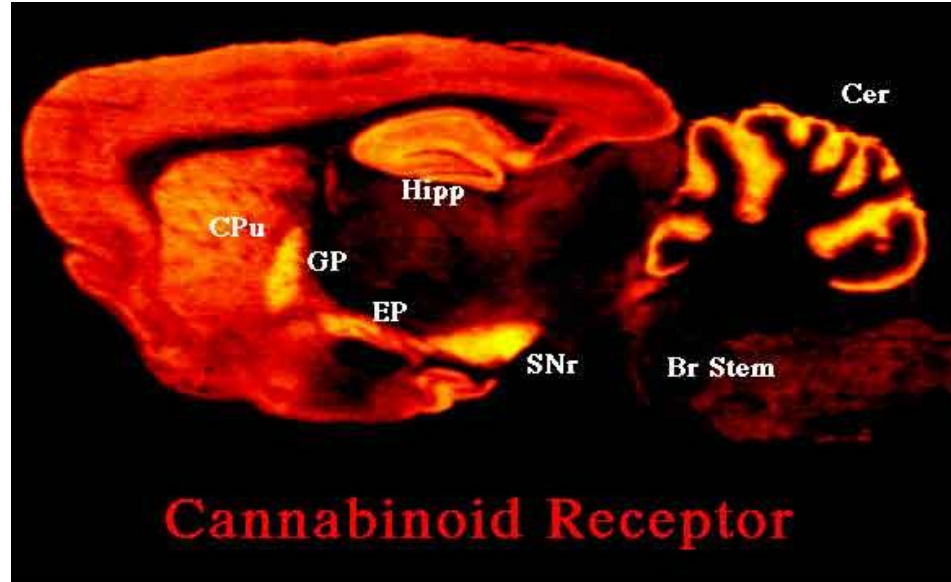
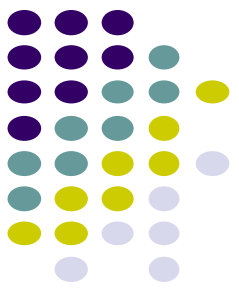




Mode of Cannabis Action

- Two specific cannabinoid receptors
 - CB1 - brain and peripheral tissues
 - CB2 - immune system
- Linked to G-proteins with diverse downstream signalling pathways
- Euphoria results from stimulation of mesolimbic dopaminergic neurons, like other drugs of abuse

Cannabinoid receptor (CB1) distribution

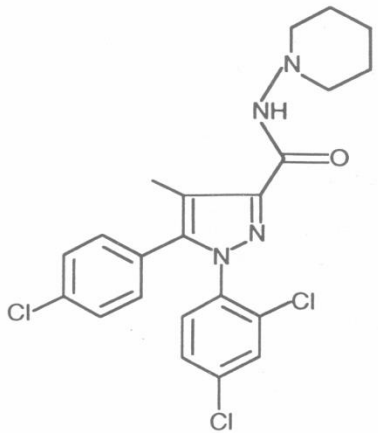


Cannabinoid receptors are localized in areas that control movement (basal ganglia, cerebellum), cognition (cerebral cortex), and attention and memory (hippocampus).

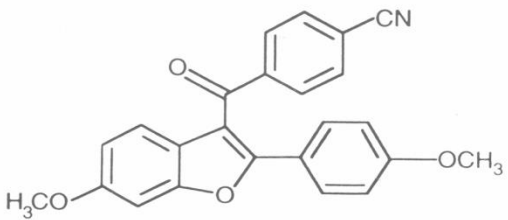
Cannabinoid receptors are notably sparse in areas that control heart rate and respiration (medulla), explaining why there are not fatal overdoses of marijuana.

Cannabinoid receptors are also located in areas that control emesis (nucleus of the solitary tract) and pain, suggesting medical potential of marijuana.

Cannabinoid receptor antagonists

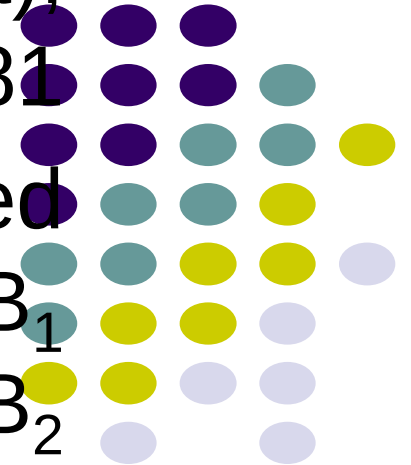


SR141716A



LY320135

SR 141716
(Rimonabant),
LY320135 and AM281
All have marked
selectivity for CB₁
receptors over CB₂





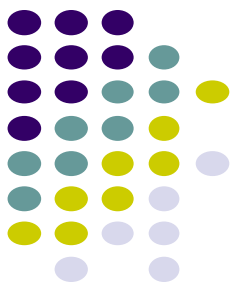
Metabolism

- rapid initial drop due to redistribution to fats
- slower metabolism in liver
- metabolites may persist for a week

Major biologically active compound may be metabolite

1. Primary metabolic product of Δ^9 -THC
(11-OH- Δ^9 -THC) is more potent than Δ^9 -THC
2. Delay between peak plasma levels and “high”

Therapeutic use of Cannabinoids



Promising indications:

- controlling nausea and vomiting (cancer, HIV)
- stimulating appetite (cancer, HIV)
- slight spasticity with pain (multiple sclerosis)
- (chronic) pain
- Gilles de la Tourette's syndrome
- asthma
- extra pyramidal symptoms
- therapy resistant glaucoma

Registered products

Dronabinol (Marinol®)

- For severe nausea and vomiting associated with cancer chemotherapy.
- For the treatment of AIDS-related anorexia

Nabilone (Cesamet®)

- For severe nausea and vomiting associated with cancer chemotherapy





Potential medical uses

- Glaucoma (increased intraocular pressure)
- Antiemetic (reduce nausea and vomiting)
- Anticonvulsant
- Enhance appetite (e.g., AIDS patients)
- Analgesic